

Uncertainty Estimation with Limited Data through Graph-Based Information Sharing

Author^{*}, *Member, IEEE*, Author, Author,
Author, and Author, *Senior Member, IEEE*

Kansas State University
Manhattan, Kansas, USA

{author1, author2, author3, author4, author5}@ksu.edu

^{*}Corresponding author

Abstract—Prediction intervals describe how far observed values may differ from forecasts. Estimating these intervals is challenging when a location has limited recent data. We propose a variance-weighted graph-smoothing method that shares information between connected locations to estimate location-specific scales. We weight the estimates using their estimated log-scale variance, and a separate calibration step adjusts interval widths toward the chosen coverage target. We evaluate the method using STGCN speed predictions on two datasets, METR-LA and PEMS-BAY, varying the data available for interval estimation at one target sensor at a time. With a short recent history, graph smoothing lowers mean Winkler scores relative to separate sensor-specific scales, while its benefit over pooling data from other sensors varies between networks. With the target sensor’s full available data for interval estimation, mean scores are close to those from separate sensor-specific scales. A second application to COVID-19 hospital admissions shows lower mean scaled Winkler scores than pooling data from other states when the target state has limited data for interval estimation. The method extends interval estimation to limited-data locations by combining graph-based scale sharing with variance weighting and calibration.

Index Terms—Machine learning, conformal prediction, graph smoothing, spatio-temporal data, uncertainty quantification, traffic forecasting, epidemic forecasting.

I. INTRODUCTION

A point forecast gives one estimate of a future observation, while a prediction interval describes how far the observation may differ from that estimate. Traffic observations form graph-structured time series, with a speed sequence at each sensor. Graph-based models predict speeds by capturing spatio-temporal dependencies across sensors and time [1], [2]. These predictions are uncertain because traffic conditions vary and models learn from limited data [3], [4].

Estimating a prediction interval becomes difficult when a target location has limited recent data for interval estimation. In traffic networks, this can arise after a sensor is activated or when its earlier observations are unavailable. We use cold start to describe having no past data available for interval estimation. Changing conditions can also make recent observations more relevant than older ones [5]. A scale estimated from a short period can be sensitive to which observations are available. Connected locations can provide additional information [6], but forecast errors, the differences between observed and predicted values, can have different magnitudes across

locations. Sharing information therefore requires a balance between the target location’s data and data from its neighbors.

Our contribution is a post-hoc variance-weighted graph smoother for location-specific uncertainty scales under limited recent data. CoRel [6] and SCALE [7] learn uncertainty models, while GraphLCP [8] uses graph structure to weight calibration samples. We instead smooth log-scale estimates over a supplied graph, giving greater weight to estimates with lower estimated variance. This lets connected locations inform the target scale when its own estimate is unreliable or unavailable. A separate split-conformal quantile rule calibrates interval widths [9]. We evaluate how Winkler scores change as the target location’s available data decreases.

Our main evaluation uses STGCN-based traffic forecasts [1] on METR-LA and PEMS-BAY [2]. In Los Angeles, using the target sensor’s latest 12, 24, or 48 five-minute slots, Graph Smoothing reduces mean Winkler score [10] by 3.3–6.6% compared with directly pooling data from the other sensors. In Bay Area, it lowers the score by 6.7–16.4% compared with using separate sensor-specific scales, with scores comparable to pooling. Having access to the target sensor’s full available data for interval estimation brings the mean scores of Graph Smoothing and separate sensor-specific scales close together. We also apply the method to forecasts of COVID-19 hospital admissions across U.S. states, using data from the COVIDcast repository [11]. In this second application, graph smoothing lowers mean scaled Winkler scores relative to pooling data from the other states when the target state has limited data for interval estimation.

The remainder of this paper is organized as follows. Section II reviews related work, followed by the proposed method in Section III. Section IV describes the experimental setup for both applications, while Section V presents and discusses their results. Section VI concludes the paper.

II. RELATED WORK

We review related approaches to prediction intervals for time series and graph data. Split conformal prediction fits a model on one part of the data and uses another part to set interval widths. Keeping calibration separate from model fitting supports its marginal coverage guarantee under exchangeability [9]. Lei et al. [9] also study a locally weighted

extension that scales absolute prediction errors before calibration, allowing interval widths to vary across inputs. Romano et al. [12] develop CQR, which calibrates lower and upper quantile predictions. Both approaches adapt interval widths to differences across inputs.

Time-series methods address dependence and distribution shift through sequential updates. Gibbs and Candès [5] introduce ACI, which adjusts the calibration level according to whether recent intervals contain the observations. Xu and Xie [13] develop EnbPI, which combines bootstrap ensemble predictions with an updated collection of prediction errors. Their analysis bounds coverage gaps under assumptions on temporal dependence and estimation accuracy. These methods use new observations to update intervals over time.

Graph-based methods use relationships between nodes when constructing prediction sets. Lunde et al. [14] establish conformal validity for regression with network-derived features under joint exchangeability assumptions. Baghershahi et al. [8] propose GraphLCP, which uses graph connections and node features to weight calibration samples through a Personalized PageRank kernel. For correlated time series, Cini et al. [6] introduce CoRel, which learns relationships between time series and uses an STGNN to estimate prediction-error quantiles. Guo et al. [7] propose SCALE, which combines graph-wavelet decomposition with a learned quantile model and evaluates it on traffic datasets. Limmer [15] studies filtered conformal ellipsoids, using predictive covariances to construct joint prediction regions for multiple sensors.

Conformal prediction has also been used to estimate uncertainty in epidemic forecasts. Wohlfender et al. [16] apply ACI to construct prediction intervals for machine learning forecasts of COVID-19 hospital admissions. For evaluating epidemic forecasts, Bracher et al. [17] describe interval scores that account for both interval width and observations outside the intervals. These works address interval construction and evaluation in the hospital-admissions application considered here.

Our work focuses on interval estimation when one target location has little or no recent data while other locations retain their available data. CoRel [6] and SCALE [7] learn models for uncertainty estimation, while GraphLCP [8] weights calibration samples using graph structure. Our method instead smooths location-specific log-scales over a supplied graph, with weights based on their estimated variance. This is post-hoc interval estimation, applied to existing forecasts. We evaluate how this scale sharing affects prediction intervals as the target location's available data changes.

III. GRAPH-SMOOTHED PREDICTION INTERVALS

Let $V = \{1, \dots, p\}$ be the nodes of a graph, with observations y_{jt} and forecasts $\hat{y}_{jt}$ at node $j \in V$ and time t . For target node $i \in V$, radius r_i defines

$$I_{it} = [\hat{y}_{it} - r_i, \hat{y}_{it} + r_i]. \quad (1)$$

For a chosen miscoverage level $\alpha \in (0, 1)$, we target coverage $1 - \alpha$ separately for each forecast horizon and amount of target-node data.

A. Starting Scales

For usable estimation times $\mathcal{E}_j$ at node j and count $n_j = |\mathcal{E}_j|$, define scaled absolute forecast errors

$$d_{jt} = \frac{|y_{jt} - \hat{y}_{jt}|}{\tau_j}, \quad \tau_j > 0. \quad (2)$$

The normalization factor τ_j is fixed before interval estimation and remains available at cold start.

For sorted values $x_{(1)} \leq \dots \leq x_{(n)}$, define $Q_{1-\alpha}(x) = x_{(1 + \lceil (1-\alpha)(n-1) \rceil)}$. With floor $\epsilon = 10^{-6}$, the pooled fallback and starting scales are

$$\begin{aligned} s_{-i} &= \max\{\epsilon, Q_{1-\alpha}(\{d_{jt} : j \neq i, t \in \mathcal{E}_j\})\}, \\ s_j &= \begin{cases} \max\{\epsilon, Q_{1-\alpha}(\{d_{jt} : t \in \mathcal{E}_j\})\}, & n_j > 0, \\ s_{-i}, & n_j = 0. \end{cases} \end{aligned} \quad (3)$$

From an older period of length T , sample B consecutive windows of length $\ell = \min(m, \lfloor T/2 \rfloor)$ for reference length $m \leq T$, with starts drawn uniformly with replacement. Let $\mathcal{E}_{jb}$ contain the usable times for node j in window b , with $n_{jb} = |\mathcal{E}_{jb}|$. For $n_{jb} > 0$, define

$$u_{jb} = \log[\max\{\epsilon, Q_{1-\alpha}(\{d_{jt} : t \in \mathcal{E}_{jb}\})\}].$$

Using these nonempty windows, we estimate

$$\begin{aligned} \hat{v}_j &= \frac{\ell}{m} \max\{10^{-8}, \text{Var}_b(u_{jb})\}, \\ c_j &= n_j^{\text{ref}} \hat{v}_j, \quad w_j = \frac{n_j}{c_j}. \end{aligned} \quad (4)$$

Here Var_b is the sample variance across usable windows. For $m < T$, $n_j^{\text{ref}} = \text{round}[(m/\ell) \text{median}_b n_{jb}]$. For $m = T$, it counts all usable observations in the older period. Dependent-quantile asymptotics motivate the approximation $\text{Var}(\log \hat{s}_j) \approx c_j/n_j$ and inverse-variance weights [18]. A limited-data target uses $\text{median}_{j \neq i} c_j$ over finite positive coefficients in place of c_i . We set $w_j = 0$ when $n_j = 0$ or variance is unavailable.

B. Graph Smoothing

Let $A = (a_{jk})$ be the symmetric weighted adjacency matrix, $E = \{\{j, k\} : j < k, a_{jk} > 0\}$, and $\mathcal{O} = \{j \in V : w_j > 0\}$. For $\lambda \geq 0$, weighted graph Tikhonov regularization [19] gives

$$\begin{aligned} \hat{\mathbf{z}} \in \arg \min_{\mathbf{z}} \Big\{ & \sum_{j \in \mathcal{O}} w_j (z_j - \log s_j)^2 \\ & + \lambda \sum_{\{j, k\} \in E} a_{jk} (z_j - z_k)^2 \Big\}. \end{aligned} \quad (5)$$

Weighted least squares gives more influence to estimates with lower estimated variance, while Laplacian regularization penalizes differences between connected log-scales. The convex

quadratic objective yields a linear system, with $W = \text{diag}(w_j)$ and graph Laplacian $L = \text{diag}(A\mathbf{1}) - A$,

$$(W + \lambda L)\hat{\mathbf{z}} = W \log \mathbf{s}, \quad \tilde{s}_j = e^{\hat{z}_j} > 0. \quad (6)$$

Log-space smoothing penalizes relative differences through $\hat{z}_j - \hat{z}_k = \log(\tilde{s}_j/\tilde{s}_k)$.

For $\lambda > 0$, each connected component intersecting $\mathcal{O}$ has a unique solution to (6). Within these components, nodes outside $\mathcal{O}$ receive a harmonic extension, a weighted average of neighboring log-scales. Components disjoint from $\mathcal{O}$ use s_{-i} . At $\lambda = 0$, we define $\tilde{s}_j = s_j$ for $j \in \mathcal{O}$ and adopt the fallback convention $\tilde{s}_j = s_{-i}$ otherwise.

The degrees-of-freedom ratio is the normalized trace of the linear smoother,

$$\rho(\lambda) = \frac{\text{tr}[(W + \lambda L)^{-1}W]}{|\mathcal{O}|}, \quad \lambda > 0. \quad (7)$$

Matrices are restricted to components intersecting $\mathcal{O}$, with $\rho(0) = 1$. Smaller ratios give stronger sharing. Tuning selects one ratio, and each target case’s weights determine λ . A special grid entry labeled zero uses a large finite λ to approximate the component-wise shared-scale limit.

C. Calibration and Comparisons

Let $\mathcal{C}$ index usable node–time calibration pairs and $\mathcal{C}_i = \{(j, t) \in \mathcal{C} : j \neq i\}$. For N sorted scores, the split-conformal rule [9] gives $Q_{\text{cal}}(x) = x_{(\lceil (1-\alpha)(N+1) \rceil)}$, with $x_{(N+1)} = +\infty$.

Set $b_j = 1$ for Global, $b_j = s_j$ for Local, and $b_j = \tilde{s}_j$ for Graph Smoothing. The nonconformity scores d_{jt}/b_j measure scale-adjusted forecast errors. Each method calculates

$$\begin{aligned} q_i &= Q_{\text{cal}}(\{d_{jt}/b_j : (j, t) \in \mathcal{C}_i\}), \\ r_i &= \tau_i q_i b_i. \end{aligned} \quad (8)$$

The multiplier q_i adjusts interval size toward coverage $1 - \alpha$. Local provides an unsmoothed reference under the same calibration procedure. Heterogeneity across nodes and temporal dependence can shift coverage from the target. We therefore report empirical coverage on test data.

IV. EXPERIMENTAL SETUP

We evaluate whether Graph Smoothing improves prediction intervals with limited data. We compare Global, Local, and Graph Smoothing across different amounts of available data and forecast horizons, with $\alpha = 0.10$ in both applications.

We split data chronologically into training (60%), validation (10%), tuning (10%), calibration (10%), and testing (10%). We use application-specific tuning windows to accommodate the sampling frequency and forecast horizon. Each reference variance estimate uses 200 sampled windows. For each target history and forecast horizon, we average tuning Winkler scores equally across locations and select the least smoothing among 15 settings within 1% of the lowest mean score. Interval radii remain fixed during testing.

A. Traffic Forecasting

We use METR-LA from Los Angeles and PEMS-BAY from the San Francisco Bay Area [2], which contain traffic speeds in mph at five-minute intervals (Table I). The DCRNN graphs connect sensor nodes with weights that decrease with road-network distance [2]. For Graph Smoothing, we symmetrize each graph using the larger of the two directed weights and remove self-connections. Fig. 1 shows the Los Angeles network.

TABLE I
TRAFFIC DATASETS WITH FIVE-MINUTE OBSERVATIONS. EDGES ARE UNDIRECTED AND EXCLUDE SELF-CONNECTIONS.

Dataset	Sensors	Time points	Missing (%)	Edges
Los Angeles	207	34,272	8.109	1,313
Bay Area	325	52,116	0.003	2,079

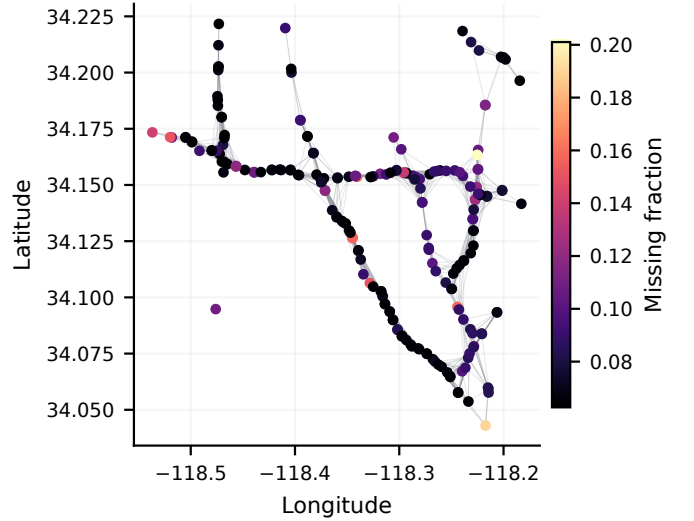


Fig. 1. Los Angeles sensor network. Node colors show the fraction of readings treated as missing.

We train an STGCN-based predictor [1] separately for each dataset. It uses 12 past speed measurements to predict the next 12 steps. We evaluate forecasts 15, 30, and 60 minutes ahead. Training uses Adam with a learning rate of 10^{-3} and a batch size of 32. We select the checkpoint with the lowest validation MAE. We treat nonfinite and nonpositive speeds as missing. Forward-fill imputation replaces missing inputs with the sensor’s last available speed [20], with its training-period median as the fallback when no earlier reading is available.

Tuning uses an approximate 60/20/20 split for older scale estimates, calibration and recent scales, and candidate scoring, with 12-slot boundary offsets. Among observations available before scoring, the middle portion’s latest 48 slots estimate recent scales and earlier slots calibrate intervals. With $\tau_j = 1$, each target sensor uses 0, 12, 24, or 48 recent five-minute slots, or its full available data. Other sensors retain their full available data.

For testing, full data combines tuning data with the latest 48 calibration slots available before testing begins. Limited-data

targets use their assigned recent slots. Earlier calibration data from other sensors determines interval multipliers. Evaluation requires an observed target speed and 12 observed input measurements at the target sensor.

B. COVID-19 Hospital Admissions

We use daily confirmed COVID-19 hospital admissions from HHS through COVIDcast [11], distributed with the geographic graph at <https://github.com/Alistair-Turcan/EpiPatch>. The data span 978 days from August 23, 2021, to April 26, 2024, with $y_{jt} \geq 0$ and no missing values. The largest connected component covers the 48 contiguous states and Washington, D.C., with $|V| = 49$, $|E| = 109$, and $a_{jk} \in \{0, 1\}$.

Each node represents a state. For seven-day-ahead forecasts, we use

$$\begin{aligned} \tau_j &= \sigma_{j,\text{train}}, \\ I_{it} &= [\max(0, \hat{y}_{it} - r_i), \hat{y}_{it} + r_i], \end{aligned} \quad (9)$$

where $\sigma_{j,\text{train}}$ is the state's training-period standard deviation. Each state serves as target i with history $k \in \{0, 7, 14, \text{full}\}$. Let $\mathcal{T}$ contain the 92 tuning dates and $\mathcal{R}_k$ the latest k available calibration dates, with $\mathcal{R}_0 = \emptyset$. The estimation sets are

$$\mathcal{E}_j = \begin{cases} \mathcal{R}_k, & j = i, k \in \{0, 7, 14\}, \\ \mathcal{T} \cup \mathcal{R}_{14}, & \text{otherwise.} \end{cases} \quad (10)$$

Tuning uses the first 32 dates for older scale estimates and the final 21 for scoring. Among intervening observations available before scoring, the latest 14 dates estimate recent scales and earlier dates calibrate intervals. For testing, the earlier 78 calibration dates from states $j \neq i$ determine q_i . Testing covers 92 daily forecasts per state from January 26 to April 26, 2024.

C. Evaluation Metrics

We compare the three methods at a 90% coverage target using Winkler score, empirical coverage, and interval width. Winkler score penalizes wide intervals and observations outside them, with lower scores indicating better results [10], [17]. Empirical coverage is the fraction of test observations inside the intervals. For traffic, both Winkler score and interval width are reported in mph. For hospital admissions, we divide each state's Winkler score by its training-period standard deviation τ_j to give states with different admission levels comparable influence, while interval width is reported in admissions. Within each application, we average each metric within each location and then equally across locations, separately for each forecast horizon and amount of available data.

Code and instructions for reproducing the experiments are available at <https://github.com/Arashlf/graph-smoothed-intervals-mlldb>.

V. RESULTS AND DISCUSSION

We examine how the three approaches compare as the amount of available data changes.

A. Traffic Forecasting Results

Table II compares scores, coverage, and widths for forecasts 30 minutes ahead using the target sensor's latest 24 five-minute slots. Fig. 2 shows the broader comparison in Los Angeles. When we use data from the target sensor's latest 12, 24, or 48 five-minute slots, Graph Smoothing lowers mean Winkler score by 3.3–6.6% relative to Global and 14.9–22.1% relative to Local. These ranges cover the three amounts of available data and all forecast horizons. When we use no data from the target sensor to estimate its interval, Graph Smoothing lowers the score by 3.5–7.0% relative to Global.

TABLE II
TEST RESULTS FOR FORECASTS 30 MINUTES AHEAD USING THE TARGET SENSOR'S LATEST 24 FIVE-MINUTE SLOTS. VALUES ARE SENSOR AVERAGES, WITH A 90% COVERAGE TARGET. BOLD MARKS THE LOWEST MEAN WINKLER SCORE IN EACH DATASET.

Method	Winkler score (mph)	Coverage (%)	Width (mph)
<i>Los Angeles</i>			
Global	36.19	90.14	16.50
Local	43.93	80.53	15.94
Graph Smoothing	34.66	89.82	15.88
<i>Bay Area</i>			
Global	21.64	88.27	7.06
Local	23.59	78.64	6.42
Graph Smoothing	21.51	87.17	6.77

In Bay Area, Graph Smoothing lowers mean Winkler score by 6.7–16.4% relative to Local when we use data from the target sensor's latest 12, 24, or 48 five-minute slots. Its scores are close to Global, differing by less than 1.3%. When we use no data from the target sensor to estimate its interval, Graph Smoothing gives the same intervals as Local.

Fig. 3 shows how the score differences vary across Los Angeles sensors. In the cases using 0, 12, 24, or 48 five-minute slots for the target sensor, Graph Smoothing gives lower scores than Global for about 75–80% of sensors. In Bay Area, it gives lower scores for about 50–58% of sensors in the cases using 12, 24, or 48 slots, and the median reductions remain near zero. These results show that the average score reductions are shared by more sensors in Los Angeles than in Bay Area.

When we use the target sensor's full available data, Graph Smoothing gives the same intervals as Local for all three Bay Area forecast horizons and for 15- and 30-minute Los Angeles forecasts. For 60-minute Los Angeles forecasts, their mean Winkler scores differ by about 0.1%, with Local slightly lower. At zero target history, Local uses a pooled starting scale from other sensors while retaining scale-normalized calibration. Its mean scores are close to Global's, with slightly different intervals. With a short history, Local replaces this pooled fallback with a scale estimated from only a few target observations. That estimate can be unstable, which helps explain why using a short history can perform worse than using the pooled fallback.

Local substantially undercovers with limited target data. For 30-minute forecasts using 24 recent slots, its mean coverage is 80.5% in Los Angeles and 78.6% in Bay Area, roughly 10 percentage points below the 90% target. Graph Smoothing

Lower is better

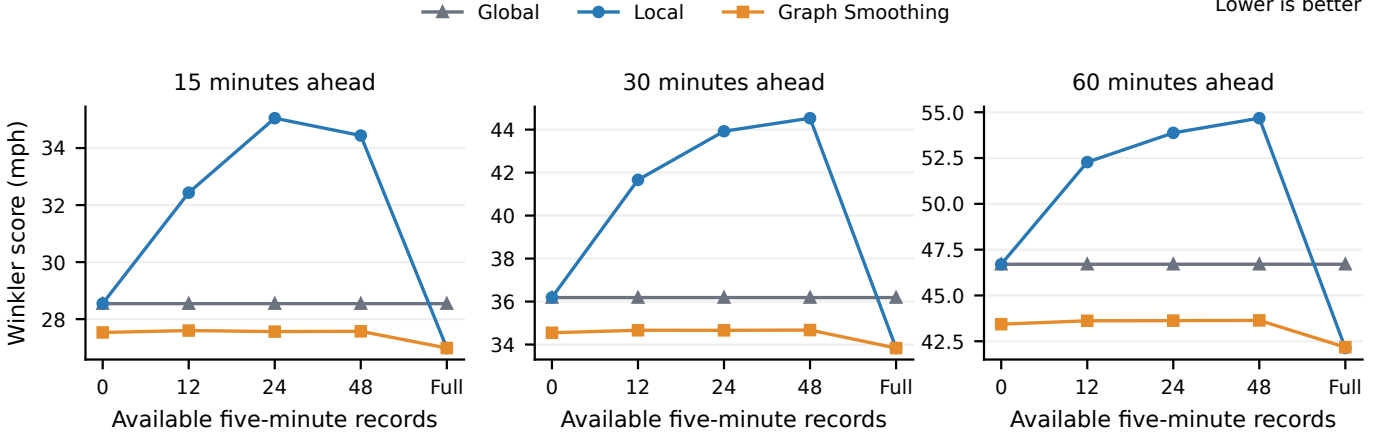


Fig. 2. Mean Winkler score in Los Angeles. We vary the amount of data used to estimate the target sensor’s interval. Other sensors retain their full available data. Full uses all data available for interval estimation. Local and Graph Smoothing have nearly identical mean scores at Full.

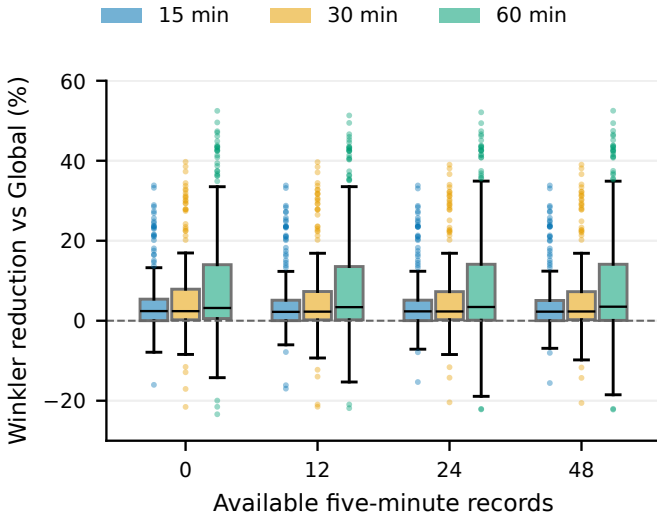


Fig. 3. Sensor-level percentage reductions in Winkler score relative to Global in Los Angeles. Positive values favor Graph Smoothing. Boxes span the interquartile range (IQR) of sensor-level reductions, with a line at the median.

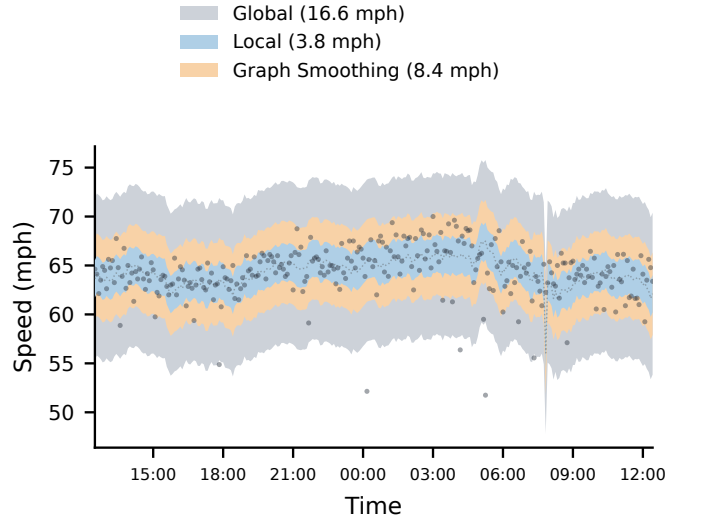


Fig. 4. Intervals for a Los Angeles sensor using data from its latest 12 five-minute slots and forecasts 30 minutes ahead. This example illustrates a case where Graph Smoothing improves on both alternatives. Bands show prediction intervals, and dots show observed speeds.

raises coverage to 89.8% and 87.2%, respectively, bringing both substantially closer to the target.

Fig. 4 shows the intervals for one Los Angeles sensor. Local produces a narrow interval that misses many observed speeds, while Global produces a much wider interval. The Graph Smoothing interval is wider than Local and narrower than Global. We selected this sensor as an example, while the results above summarize all sensors.

B. COVID-19 Hospital Admissions Results

Graph Smoothing also lowers mean scaled Winkler scores in the hospital-admissions application. Fig. 5 compares the three approaches as the target state’s available data changes. With 0, 7, and 14 days of target data, Graph Smoothing reduces mean scaled Winkler score by 10.2%, 5.6%, and 3.1% relative to Global, respectively. It also gives lower mean scores than

Local in each of these cases. With full available data, Graph Smoothing and Local give the same intervals.

With limited target data, Graph Smoothing gives mean interval widths of 60.5–64.0 admissions, compared with 74.5 for Global. Although narrower than Global’s, these intervals have mean empirical coverage of 96.7–98.7%, above the nominal 90% target. This application demonstrates the use of the same scale-sharing method around hospital-admission forecasts.

C. Discussion and Limitations

Graph-based scale sharing offers a practical approach to interval estimation with limited local data. Its gains vary across datasets, and our evaluation covers one test period per dataset. The comparisons assess the complete method,

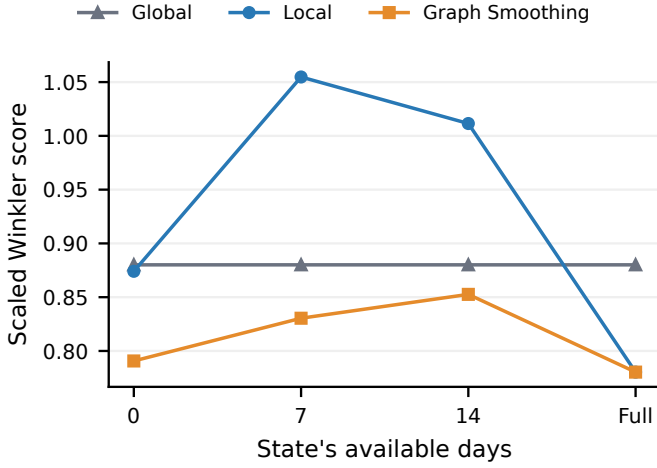


Fig. 5. Mean scaled Winkler score for seven-day-ahead hospital-admission forecasts. Each state's score is divided by its training-period standard deviation, then scores are averaged equally across locations. Local and Graph Smoothing overlap at Full.

leaving the separate contributions of variance weighting and graph structure to be quantified. Empirical coverage also varies around the nominal target.

Additional temporal periods could show how score differences change over time. Paired comparisons could add uncertainty estimates for these differences while accounting for temporal dependence. Ablations with count-only weights and graph-free shrinkage could quantify the contributions of weighting and connectivity. Node-level coverage analysis could show how coverage varies across locations.

Future work can explore privacy-preserving scale sharing, online calibration under distribution shift, learned or time-varying graphs, and joint prediction intervals across multiple locations. These directions could extend uncertainty estimation to networks with changing relationships and restricted data access.

VI. CONCLUSION

Prediction intervals express uncertainty alongside point forecasts, including at locations with limited recent data. We presented a method that combines variance-weighted graph smoothing of location-specific log-scales with calibration. In traffic forecasting and hospital admissions, the method lowers mean Winkler scores relative to separate location-specific scales when target histories are short, with gains over pooling that vary across datasets. This approach uses information from connected locations to support interval estimation when local data are scarce.

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